

Week5_Diagnostics_Report_CharleneNyakio

Operational Diagnostics Report

Context

The objective of this investigation was to identify the operational factors contributing to reduced system performance using the Mystery Ops Dataset. The analysis followed a structured diagnostic workflow consisting of data profiling, anomaly detection, drill-down analysis, Pareto analysis and correlation analysis. The dataset contained approximately 2,920 operational records with only a few missing values, indicating that it was suitable for analysis after basic data quality checks.

Insight

Data Profiling: Summary statistics and missing-value analysis showed only minor gaps (18 Operator IDs, 5 Throughput values and 20 Temperature values). Distribution plots indicated that throughput was not normally distributed and contained several extreme observations.

Anomaly Detection: Box plots highlighted unusually low throughput observations together with voltage outliers. These anomalies aligned with operational maintenance events, suggesting that they represented genuine operational issues rather than data errors.

Root Cause Analysis: Drill-down analysis by depot identified Nairobi as the weakest-performing location, recording the lowest average throughput while also experiencing the highest maintenance activity. Pareto analysis showed that missed maintenance accounted for roughly 90% of recorded incidents (529 out of 588 incidents), demonstrating that a single issue was responsible for the majority of operational losses. Correlation analysis further confirmed a strong negative relationship ($r \approx -0.85$) between maintenance activity and

Key Finding	Evidence
Main Issue	Missed maintenance
Highest Impact	$\approx 90\%$ of incidents linked to missed maintenance
Affected Area	Nairobi depot
Evidence	Strong negative correlation between maintenance and throughput ($r \approx -0.85$)

throughput, while temperature and voltage had much weaker relationships.

≈ -0.85)

Action

1. Prioritize preventive maintenance schedules to reduce equipment downtime.
2. Implement automated monitoring and alerts for abnormal throughput and overdue maintenance tasks.
3. Monitor maintenance compliance and throughput KPIs monthly, with particular attention to the Nairobi depot, to measure operational improvement over time.

Conclusion

The operational diagnostics indicate that maintenance practices are the primary driver of reduced operational performance. Because a single issue contributes to the majority of incidents, targeted maintenance improvements are expected to deliver the greatest operational benefit. This report is written for operational decision-making and supports prioritizing preventive maintenance over broad system-wide interventions.